

Project Title:**Brute-Force Attack Simulation on SSH and RDP Services Using Hydra and Metasploit in a Dockerized Lab**

Topic:

Simulating and demonstrating brute-force attacks against both SSH and RDP services within a controlled Docker environment to showcase the dangers of weak credentials and the critical importance of system hardening and patching.

End Goal / Vulnerability Being Exploited:

To exploit weak SSH and RDP credentials (specifically the commonly used password **toor**) through automated brute-force attacks using Hydra. For the Windows RDP target, a secondary goal is to scan for the **BlueKeep vulnerability (CVE-2019-0708)** using Metasploit. This project highlights how attackers can compromise poorly secured services and why patch management is vital for legacy vulnerabilities.

Devices and Technologies Used:

- **Docker & Docker Compose** – To isolate and orchestrate attacker and target containers in a repeatable environment.
- **Kali Linux Container** – Used as the attacking machine, equipped with Hydra, Nmap, Metasploit, and supporting tools.
- **Ubuntu-based SSH Target Container** – Hosts an SSH service with a weak password.
- **Windows-based RDP Target Container** – Simulates a Windows environment running Remote Desktop Protocol (RDP) with weak credentials and a vulnerable configuration.
- **Ansible** – Automates setup and provisioning of all lab containers and required tools.
- **Python** - scripts would be launch from Kali machine to Linux and Windows server containers
- **Hydra** – Used to brute-force both SSH and RDP login credentials.
- **Metasploit** – Used to scan the Windows container for the BlueKeep RDP vulnerability.
- **Colima (on macOS)** – Provides Docker backend support on macOS.

Summary of Implementation:

1. **A custom Ansible playbook launches three Docker containers: a Kali Linux attacker, an Ubuntu SSH target, and a Windows RDP target.**

2. All containers are attached to a shared Docker network (**pentest-net**) for internal testing.
3. The SSH container exposes port **22** with a default user configured with a weak password (**toor**).

The Windows container exposes port **3389** (RDP) and is preloaded with similarly weak credentials (**Administrator:toor**).

4. From the Kali container:
 - Nmap is used to scan for active hosts and confirm open ports (22 for SSH, 3389 for RDP).
 - Hydra performs brute-force attacks on both SSH and RDP services using a password list.
 - A Metasploit auxiliary scanner checks the Windows target for the BlueKeep vulnerability.
5. Attack results (successful brute-force login and vulnerability presence) are logged or printed to the console, emphasizing the ease with which attackers can compromise poorly secured and unpatched systems.